Ethan Buckley

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Natural Sciences undergraduate (Physics major) at UCL, working as a data science intern applying Bayesian statistics and machine learning to live research data. Hierarchical models fitted with MCMC, and the Python pipelines that keep them reproducible. Seeking data science and quantitative research internships.

EDUCATION

University College London, MSci Natural Sciences — Sep 2024 – Jun 2028

Physics major, Physical Chemistry minor. Modules include Programming for Scientists, Chemical Dynamics, Linear Algebra and Vector Calculus, Chemistry of Materials, Mathematical Methods III, Quantum Physics, Atomic and Molecular Physics, Thermal Physics, Mathematics for Natural Sciences.

Queen Elizabeth School, Kirkby Lonsdale — 2017 – 2024

A-levels: Mathematics, Chemistry, Biology, Physics; awarded school Physics prizes (GCSE and Year 12). GCSEs: ten subjects including Mathematics, Physics and Computer Science.

EXPERIENCE

Data Science & Research Intern, Down Syndrome Education International — Jun 2026 – Present

- Build and extend hierarchical Bayesian models (PyMC, MCMC) of vocabulary growth in children with Down syndrome, using Beta-Binomial likelihoods with age-varying dispersion and Hilbert-space Gaussian-process trajectories, fitted to Communicative Development Inventory data pooled across UK, US and Italian cohorts plus around 6,100 typically-developing children from Wordbank.
- Contribute to a Bayesian re-analysis of a randomised controlled reading-intervention trial (54 children, four assessment waves), using LightGBM/SHAP predictor ranking and a DAG-based causal framework for treatment-effect and mediation analysis.
- Rebuilt a fragile single-notebook pipeline into modular, auditable per-source ETL scripts and replaced a hard-coded anonymisation key with environment-based secrets; work in Python with DuckDB, Quarto, Azure Blob Storage and Git/GitHub review.
- Build the pipeline that generates DSE's teaching images and audio from a written content specification. Its main rule is contrast, for the reduced contrast sensitivity common in Down syndrome.
- Chose the voices by measurement rather than preference, using a blind A/B listening arena with Bradley–Terry ranking and bootstrap confidence intervals across four commercial text-to-speech providers (4,307 clips, 10 English accents), and halved the image unit cost with a verified batch-inference integration.

UCL Centre for Computational Science, Supervised Literature Review — 2026

- Selected to review quantum-GPU computing and variational quantum algorithms for biomolecular simulation, under Professor Peter Coveney FRS.

Seasonal Team Member, Roadchef, Killington Lake — Jul 2025 – Apr 2026

- Handled high-volume transactions and inventory under pressure; diagnosed and resolved faults in site-critical equipment to minimise downtime.

Nuclear industry placements, BAE Systems (Barrow) & EDF Energy (Heysham 1) — Jul–Aug 2023

- Placements across engineering, chemistry and maintenance in safety-critical nuclear environments, including reactor-chemistry monitoring in the Heysham laboratory.

TECHNICAL SKILLS

Programming: Python (NumPy, pandas, PyMC, LightGBM, SHAP, Matplotlib, scikit-learn, pytest), SQL / DuckDB, Git / GitHub, LaTeX, Quarto, generative-model APIs (batch inference).

Statistics & ML: Bayesian inference and MCMC, hierarchical / random-effects models, Gaussian processes, gradient boosting, causal inference (DAGs), blind A/B evaluation with Bradley–Terry ranking and bootstrap intervals, hypothesis testing, error analysis.

PROJECTS

- Algorithmic S&P 500 stock screener (independent): combined an XGBoost classifier with a FinBERT sentiment model, trained on around 1.3 million daily observations. Built 22 technical indicators (RSI, MACD, VWAP, ATR) and used Triple-Barrier-Method labelling to avoid look-ahead bias; validated with expanding-window walk-forward testing over 1,630 out-of-sample days (2020–2026): top-15 precision around 48% versus a 29% base rate.
- SEIQR epidemic cellular automaton (extended from a UCL group project): vectorised the simulation with SciPy convolution and NumPy masking for a roughly 40× speed-up, and validated it against its analytical mean-field SEIQR ODE limit. Reproducible from fixed seeds, with bit-identical regression tests in CI on Python 3.10–3.12.

ADDITIONAL

UCL Men's Hockey Club and volunteer junior coach at Kirkby Lonsdale HC; Duke of Edinburgh Gold Award. Societies: AI, Natural Sciences. Interests: mountaineering, skiing, running.